METRIC

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25 November 2008
SUPERSEDES
MIL-STD-867B
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17 July 2008

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

TEMPER ETCH INSPECTION

Reactivated after 25 November 2008 and may be used for new and existing designs and acquisitions.

AMSC: N/A

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FOREWORD

1. This military standard is approved for use by 309MXSG/MXRL, Department of the Air Force, and is available for use by all departments and agencies of the Department of Defense.

2. This standard provides guidance on the processes for the grinding of chrome plated steel and steel parts for the Air Force repair, acquisition, and manufacture of parts and/or spare parts on the landing gear of all military aircraft.

3. Beneficial comments, recommendations, additions, deletions, and any pertinent data, which may be of use in improving this document, should be addressed to 309MXSG/MXRIL, Hill AFB, UT 84056-2609 or 309MXSG/MXRL@hill.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1. SCOPE

1.1 Scope. This standard establishes requirements and test for performing temper etch inspection of carburized steels and tool and low alloy steels, which are heat treated above 1.24 x 10^9 Pascal (PA) (1.26 x 10^2 kilogram – force/square millimeter (Kgf/mm^2) or 1.8 x 10^5 pound – force/square inch (psi). This standard applies to aircraft and missiles structural and propulsion system components when MIL-HDBK-6870 (for guidance only) and/or MIL-STD-866 are reference documents in the solicitation or contract.

1.2 Groups. Temper-etch inspection is typically performed on steels from the following groups:

a. Group A: Low alloy steels:

SAE 52100	SAE 4140	SAE 4330
SAE 4340	300 M	D6AC
440C		

Carburizing Steels:

SAE 4620	9310	AMS 6260

b. Group B: Tool Steels:

H-11	M-50	M-2
H-13		

NOTE: The alloys listed are examples. The process is applicable to any alloy that would be considered part of each group.

1.2.1 Case hardened steels. The standard is not applicable to surface hardened steels produced by nitriding or carbonitriding.

1.3 Purpose. This inspection is to be conducted to determine if heat has been induced in a component, subsequent to final heat treat, by machining, grinding, or other means, which will adversely affect the required properties of that component.

1.3.1 Localized discontinuous carburization. To detect localized discontinuous carburization due to an inadvertent carburize stop-off and to determine the presence of or lack of total carburization in specified locations.

1.3.2 De-carburized surface layer. To detect a de-carburized surface layer.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are

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cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, handbooks, and commercial item descriptions. The following specifications, standards, handbooks, and commercial item descriptions form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-16173	Corrosion Preventive Compound Solvent Cutback, Cold Application

(Copies of these documents are available online at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTIONS

A-A-59105	Nitric Acid, Technical
A-A-59281	Cleaning Compound, Solvent Mixtures

(Copies of these documents are available online at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5091.)

CODE OF FEDERAL REGULATIONS (CFR)

27CFR 21.35	Formula 3-A

(Copies of this document are available online at www.access.gpo.gov/nara/cfr or from the Superintendent of Documents, U.S. Printing Office, North Capitol & “H” Streets, N.W., Washington D.C. 20402-0002.)

2.3 Non-Government standards and publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE-AMS-H-6875	Heat Treatment of Steels, Raw Materials

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SAE-ARP 1923 Qualification and Certification of Etch Inspectors

(Copies of these documents are available online at www.sae.org or from the Society of Automotive Engineers International, 400 Commonwealth Drive, Warrendale, PA 15096-1001.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM – D329 Acetone

ASTM – E1146 Acid, Muriatic, (Technical Grade) Hydrochloric Acid

ASTM – E1444 Particle Testing, Magnetic

(Copies of these documents are available at www.astm.org or ASTM International, 100 Barr Harbor Drive, P.O. Box C700, Wes Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS (Not applicable.)

4. GENERAL REQUIREMENTS

4.1 Materials and equipment.

4.1.1 Materials. The use of reclaimed materials shall be encouraged to the maximum extent possible. Materials required are as follows:

a. Ethyl alcohol (27CFR 21.35, Formula 3-A) or methyl alcohol.

b. Hydrochloric Acid (HCL), (ASTM – E1146)

c. Nitric Acid (HNO_3), (A-A-59105)

d. Sodium Hydroxide (NaOH)

e. In lieu of (d) above, any common alkali solution with a pH greater than or equal to 10 may be used.

f. Ammonium Peroxydisulfate (Ammonium Persulfate) $(NH_4)_2S_2O_8$ commercial.

4.1.2 Equipment.

a. A commercial industrial vapor degreaser or a hot alkaline cleaning system.

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b. Containers used for acid solutions shall be non-reactive to acid and alkali solutions, preferably made of polyethylene or polyvinyl chloride material.

c. Water rinse tanks shall be equipped with a constant overflow or skimming device.

d. The following auxiliary equipment shall be provided:

(1) Clock or timer with second's readout that can be read from the workstation.

(2) Light capable of insuring 2152.8 Lux (200 foot candles) at inspection level.

4.1.3 Company process specifications. Company process specifications may be prepared incorporating the applicable requirements of this standard and in addition supplying the detailed information necessary to meet or exceed this standard using the particular equipment, process, personnel and test facilities required to meet the reliability requirements of the product. Personnel training, qualification and certification procedures shall be documented.

4.1.3.1 Standardization. The company process specification shall reflect procedures and records to assure adequate Quality Assurance measures are being enforced to keep the process in control. Applicable drawings or other documents shall specify the allowable severity and concentration of allowable temper etch conditions through use of photographs or wording as to allowable grinding burn severities by color, size and/or hardness differential. The component or applicable document shall also show:

a. Process name.

b. Process step number and operation name.

c. Processing time for each step.

d. Processing temperature for each step.

e. Materials used in each processing tank.

4.2 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The supplier may utilize his own facilities or any other commercial laboratory acceptable to the customer.

4.3 Manufacturing process cycle. The temper etch process shall be used after final grinding or machining operations but prior to any processing, which may interfere with this inspection, such as plating or shot peening. The etch process will remove approximately 0.0013 millimeters (0.00005 inch) of material from the surface of the part using the process in TABLE I. Material removal using the process in TABLE II could be 0.0051 millimeters (0.0002 inches) or more. The processor will need to review the

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tolerance requirements and, if necessary, allow sufficient material remaining after machining or grinding to account for the material removed during the etch process.

4.4 Record of inspection. The results of the etch inspection shall be appropriately recorded and filed.

4.5 Qualification and certification. Before performing temper etch inspection on production parts inspection personnel shall be qualified and certified in accordance with SAE-ARP-1923.

4.5.1 Initial qualification. At qualification inspection personnel shall pass physical, written, and practical tests.

4.5.2 Re-certification. Personnel performing temper etch inspection shall be re-qualified and re-certified as specified in this section at intervals not to exceed three (3) years. Physical tests are required yearly.

4.5.3 Re-examine for qualification. Personnel examined and not meeting qualification standards must wait 30 days and show evidence of having taken suitable corrective action, additional training or self-study before re-examination.

4.5.4 Certification. Records for all qualified personnel shall be maintained and include: Date of qualification, results of physical, written and practical test, and experience as a temper etch inspector.

5. DETAILED REQUIREMENTS

5.1 Process. Temper etch inspection may be accomplished utilizing a process procedure as listed in Tables I, II, III or IV for the type of steel to be inspected. The entire surface of the feature in question shall be etched and inspected.

5.2 Cleaning. Parts shall be cleaned to a water break free condition just prior to the etch process by vapor degreasing, hot ultrasonic cleaning or water based hot alkaline cleaning. To obtain a uniform surface for optimum temper etch on rough surfaces, grit blasting or liquid honing is recommended.

5.3 Solution concentrations. The acid solution concentrations listed in the tables that follow are a percent of the concentrated solution used in the makeup. For example: Solution 1 in TABLE I is 3-5% of the concentrated nitric acid solution used. That is, 3 to 5 gallons of concentrated nitric acid diluted to 100 gallons.

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TABLE I

Table I may be used to etch low alloy steels (Group A)

	PROCESS	SOLUTION	TIME	TEMPERATURE	REMARKS
1.	Nitric Acid Etch	Nitric Acid 3% - 5% (by volume) in alcohol Nitric Acid 3% - 5% (by volume) in water	15 – 60 seconds 5 – 15 seconds	Ambient	Agitate parts when immersed. Exact time may vary to produce desired black oxide film.
2.	Rinse	Circulating water	Rinse only	Ambient	To remove acid. Total immersion required.
3.	Hydrochloric Acid Dip	Hydrochloric Acid 4% - 6% (by volume) in alcohol or water Hydrochloric Acid 1% - 3% (by volume) in alcohol	30 – 60 seconds 5 – 15 seconds	Ambient Ambient	Agitate parts when immersed to remove black oxide film and provide uniform brownish grey surface.
4.	Rinse	Running water	Rinse only	Ambient	Agitate parts when immersed.
5.	Neutralize	Any alkali solution with pH of 10 minimum	15 seconds minimum	16 – 82 ˚C (60 - 180˚F)	Agitate parts when immersed.
6.	Rinse	Circulating water	Rinse only	Ambient	To remove caustic.
*7.	Rinse	Alcohol	Rinse only	Ambient	To remove water.
8.	Inspect and evaluate per appropriate criteria.				
**9.	Oil	Rust preventative oil		Ambient	To prevent corrosion
* Operational procedure: Hot water rinse (66˚C minimum {150˚F}), followed by a dry air blast may be used in lieu of this rinse. If a hot water rinse is used in lieu of the alcohol rinse, the rinse in Step 6 may be omitted.					
Pre-cleaning, water rinse and Steps 1 thru 8 shall be one continuous operation.					
** Apply the oil after the required embrittlement relief bake. This step may be omitted if the facility can show evidence that corrosion during the queue between process steps is not an issue. This could be the case in a low humidity environment.					
Maintain all rinses at less than or equal to 350 ppm dissolved solids.					

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TABLE II

Table II may be used to acid etch tool steels (Group B)

STEP	PROCESS	SOLUTION	TIME	TEMPERATURE	REMARKS
1.	Nitric Acid Etch	Hydrochloric acid 4.0% - 6.0% (by volume) in alcohol or water	1 ½ to 3 ½ minutes	Ambient	To de-oxidize part. Agitate part during entire immersion time
2.	Rinse	Circulating water	Rinse only	Ambient	To remove acid. Total immersion required.
3.	Nitric acid	Nitric acid 2.5% to 3.5% (by volume) in alcohol or water	1 ½ to 2 ½ minutes	Ambient	Agitate parts during entire immersion time. Exact time may vary to produce desired black oxide film.
4.	Rinse	Circulating water	Rinse only	Ambient	To remove acid. Total immersion required.
5.	Hydrochloric acid. Acid dip.	Hydrochloric acid 4.0% - 6.0% (by volume) in alcohol or water	1 ½ to 3 ½ minutes	Ambient	Agitate parts during entire immersion time. To remove black oxide film and produce a uniform brownish/gray surface.
6.	Rinse	Circulating water	Rinse only	Ambient	Agitate parts when immersed
7.	Neutralize	Any alkali solution with pH of 10 minutes	15 seconds minimum	Ambient	Agitate parts when immersed
8.	Rinse	Running water	Rinse only	Ambient	To remove caustic
*9.	Rinse	Alcohol	Rinse only	Ambient	To remove water
10.	Inspect and evaluate per appropriate criteria.				
**11.	Oil	Rust preventative oil	Dip only	Ambient	To prevent corrosion.
* Optional procedure: Hot water rinse (66˚C minimum {150˚F}), followed by a dry air blast may be used in lieu of this rinse. If a hot water rinse is used in lieu of the alcohol rinse, the rinse in Step 8 may be omitted.					
Pre-cleaning, water rinse and Steps 1 thru 10 shall be one continuous operation.					
** Apply the oil after the required embrittlement relief bake. This step may be omitted if the facility can show evidence that corrosion during the queue between process steps is not an issue. This could be the case in a low humidity environment.					
Maintain all rinses at less than or equal to 350 ppm dissolved solids.					

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TABLE III

(Limited Access or Swab Etch Technique)

STEP	PROCESS	SOLUTION	TIME	TEMPERATURE	REMARKS
1.	Solvent clean	Cleaner A-A-59281 or ASTM-D329	As necessary	Ambient	Wipe surface with commercial cheese cloth saturated with solvent. Wipe dry with cheese cloth.
2.	Nitric acid etch	Nitric acid (A-A-59105) 4% to 10% (by volume) in alcohol (27CFR 21.35, Formula 3-A)	Until a black smutty etch is produced.	Ambient	Swab the surface with commercial grade cheese cloth saturated with Nitric solution. The more concentrate solution reduces time.
3.	Rinse	Acetone (ASTM-D329)	To remove nitric acid	Ambient	Rinse immediately
4.	Acid clean	Hydrochloric acid (ASTM-E1146) 6% to 10% (by volume) in water	To remove black smut.	Ambient	Swab with commercial cheese cloth saturated with HCL solution.
5.	Rinse	Clean water	To remove HCL	Ambient	Swab with commercial cheese cloth saturated with water.
6.	Neutralize	Any alkali solution with pH of 10 minimum	To neutralize HCL	Ambient	Swab with commercial cheese cloth saturated with alkali solution.
7.	Rinse	Clean water	To remove caustic	Ambient	Swab with commercial cheese cloth saturated with water.
8.	Rinse	Alcohol	To remove water	Ambient	Swab with commercial cheese cloth saturated with alcohol and wipe dry.
9.	Inspect and evaluate per appropriate criteria.				
*10.	Oil	Rust preventative MIL-PRF-16173	Swab only	Ambient	To prevent corrosion.
Steps 1 thru 9 shall be completed without interruption.					
* Apply the oil after the required embrittlement relief bake. This step may be omitted if the facility can show evidence that corrosion during the queue between process steps is not an issue. This could be the case in a low humidity environment.					

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*TABLE IV

Ammonium Peroxydisulfate (Ammonium Persulfate) Swab Etch Technique

STEP	PROCESS	SOLUTION	TIME	TEMPATURE	REMARKS
1.	Solvent clean	Acetone (ASTM-D329) or cleaner (A-A-59281)	As necessary	Ambient	Wipe surface with commercial cheese cloth saturated with solvent. Wipe dry with cheese cloth.
2.	Etch	Ammonium Peroxydisulfate 10% by weight in water.	15 – 60 seconds	Ambient	Make solution just before use. Swab with commercial cheese cloth saturated with solution.
3.	Rinse	Clear water	As required to remove etch solution	Ambient	Swab area with commercial cheese cloth saturated with water.
4.	Neutralization	Any alkali solution with a pH of 10 minimum	As required to neutralize etchant	Ambient	Swab area with commercial cheese cloth saturated with alkali solution.
5.	Rinse	Clear water	As required to remove alkali solution	Ambient	Swab with commercial cheese cloth saturated with water.
6.	Rinse	Alcohol or acetone	To remove water	Ambient	Swab with commercial cheese cloth saturated with alcohol. Wipe dry.
7.	Inspect and evaluate per appropriate criteria.				
**8.	Oil	Rust preventative (MIL-PRF-16173)	Swab only	Ambient	To prevent corrosion.
*This process may also be used as an immersion technique, provided pre-cleaning is in accordance with para 5.2. (However see NOTE 2 below.)					
Steps 1 thru 7 shall be completed without interruption. NOTE 1: No bake required after etching using this process. NOTE 2: Make solution up just before use as it becomes inactive after 72 hours.					
** This step may be omitted if the facility can show evidence that corrosion during the queue between process steps is not an issue. This could be the case in a low humidity environment.					

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5.4 Acid etches visual inspection. Inspection of etched surfaces shall be performed under light resulting in a minimum of 2153 Lux (200 foot candles) at part surface.

5.4.1 Rejection criterion. The presence of the following indications shall be cause for rejection.

a. Un-tempered martensite as evidenced by light gray to white areas surrounded by a light brown to black border.

b. Over-tempered areas as evidenced by light brown to black indications darker than surrounding areas.

c. Acceptance/rejection standards. The presence of the following indications shall be cause for review to established acceptance/rejection standards (See: 4.1.3.1).

(1) Non-carburized/de-carburized areas. Non-grinding related: These areas will appear unusually lighter in color than adjacent carburized areas.

(2) Improper carburization. Non-grinding related: These areas are isolated or complete areas of carburization not required by blueprint.

5.4.2 Acceptable color. If no damage was produced by the pre-production operations, the surface will be a uniform gray. This discoloration is not detrimental to part usage.

5.5 Baking. After the final etch, all parts processed by etch methods in TABLES I, II and III shall be baked at 191 ± 14˚C (375 ± 25˚F). Under direction from the applicable engineering organization the temperature of this bake may be increased to a maximum of 28˚C (50˚F) below the final tempering temperature of the part. The bake time shall be a minimum of four (4) hours. Such baking shall be initiated within eight (8) hours of initiation of the etch inspection process and before any plating operations. Parts processed by etch methods in TABLE IV will not require a bake. All furnaces used shall be certified to SAE-AMS-H-6875.

5.5.1 Bake temperature limitation for cold worked parts. Parts that have been cold worked (e.g. shot peen, cold rolling) shall be limited to the 191 ± 14°C (375 ± 25°F) bake temperature.

5.6 In-service quality checks.

5.6.1 Solution concentrations. Process solutions shall be analyzed at a frequency determined to assure the solutions are maintained in accordance with the limits of this specification.

5.6.2 Sensitivity. A standard test piece, which contains representative grinding burns will be processed each day that the etch is operated to verify that the etching solutions and techniques are adequate to detect grinding burns. The test piece shall be either 300M steel heat treated to 280 – 300 KSI ultimate tensile strength or 4340 steel heat treated to 260 – 280 KSI ultimate tensile strength or as specified in the contract.

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5.6.3 Magnetic particle inspection. All parts shall be magnetic particle inspected in accordance with ASTM – E1444 after baking and prior to other processing, which might hinder detection of a discontinuity.

5.6.4 Records. Historical records shall be maintained on all chemical tests and on additions made to the solution.

5.7 Re-etching. A part may be etched two or more times, if dimensions and surface finishes allows, in order distinguishing burn indications from irrelevant indications; such as, smear or smutty finishes. Aluminum oxide or garnet grit blast between re-etch is recommended.

5.8 Seeking competent advice. A metallurgist shall be consulted in instances where the results of the etching appear inconclusive and where further interpretation is required.

5.9 Hardness testing. If part size and configuration permits, hardness testing in accordance with ASTM – E18 using 15/N Rockwell test equipment may be used as an evaluation of hardness in moderate to heavy over-tempered areas. A reduction of two (2) Rockwell 15/N points or more from the areas directly adjacent to the burn is indicative of an over-stressed and crack susceptive condition and should **not** be accepted in high stress areas. In lieu of Rockwell 15/N hardness testing, 1000 gram maximum load micro-hardness testing may be employed.

5.10 Rework/salvage. If sufficient stock remains for rework within the specified dimensions, the parts may be reworked to remove indications of over-tempering or re-hardening. Parts subjected to this rework shall be re-acid etch inspected.

5.11 Acid etch removal. Discoloration from the acid etch inspection process shall be acceptable unless otherwise specified by the drawing or customer requirements. The discoloration may be removed by polishing, honing, vapor grit blasting, or electrolytic alkaline cleaning methods providing dimensional changes do not exceed drawing requirements and customer approval is obtained.

5.12 Marking. Parts which have met the etch inspection requirements shall be marked in accordance with the applicable drawing, specification, purchase order or contract. Marking, as specified, shall be applied in such manner and location as to be harmless to the part and which will preclude removal, smearing, or obliterating by subsequent handling. A symbol N, NE or AE may be used. Provision shall be made to identify the individual inspector accomplishing the temper etch inspection along with the date and time the inspection was performed. In lieu of marking the part and as allowed by contract, an approved process control document (traveler, work control document, etc.) that identifies the specific part by part number and serial number may be annotated.

5.13 Toxic and hazardous substances. This document specifies use of certain materials which have been listed in subpart Z, 29 CFR 1910 (OSHA Standards), “Toxic and Hazardous Substances.” Personnel exposure to these materials must be limited to the values specified in applicable portions of 29 CFR 1910.1000.

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard provides the step by step process and procedures for temper etch to be used during repair process and acquisition of spares for the landing gear of all military aircraft.

6.2 Subject term (key word) listing.

Equipment
Materials
Process

6.3 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate modifications generated by this change. This was done as a convenience only and the Government assumes no liability whatsoever for an inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodian:
Air Force – 70

Preparing Activity:
Air Force – 70

(Project: NDTI-2009-002)

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